Question	Which condition favour the bond formation?
Type	multiple choice
Option	Maximum attraction & maximum potential energy
Option	Minimum attraction and minimum potential energy
Option	Minimum potential energy and maximum attraction
Option	None of the above
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which one of the following elements will never obey octet rule?
Type	multiple choice
Option	Na
Option	F
Option	S
Option	H
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The electro valency of the element is equal to?
Type	multiple choice
Option	Number of electrons lost
Option	Number of electrons lost
Option	Number of electrons transferred
Option	Number of electrons lost or gained by the atom of the element during the formation of ions of ionic
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Conditions for ionic bond formation is/are?
Type	multiple choice
Option	Small cation, large anion
Option	Low IP of cation, high electron affinity of anion
Option	Large cation, small anion and less charge
Option	Less lattice energy
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Born Haber cycle is mainly used to determine?
Type	multiple choice
Option	Lattice energy
Option	Electron affinity
Option	Ionisation energy
Option	Electronegativity
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	An ionic compound $A^{+}B^{-}$ is most likely to be formed when?
Type	multiple choice

Option	Ionization energy of A is low
Option	Electron affinity of B is high
Option	Electron affinity of B is low
Option	Both (a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The pair of elements which on combination are most likely to form an ionic compound is?
Type	multiple choice
Option	Na and Ca
Option	K and O
Option	O and Cl
Option	Al and I
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	As compared to covalent compounds electrovalent compounds generally possess?
Type	multiple choice
Option	High m. p. and high b. p.
Option	Low m. p. and high b. p.
Option	Low m. p. and high b. p.
Option	High m. p. and low b. p.
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The incorrect order of lattice energy among the following set is:
Type	multiple choice
Option	AlF3 > MgF2
Option	LiN > Li2O
Option	NaCl > LiF
Option	TiC > ScN
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The lattice energy of NaCl is 788 kJ mol-1. This means that 788 kJ of energy is required:
Type	multiple choice
Option	to separate one mole of solid NaCl into one mole of Na(g) and one mole of Cl(g) to infinite distance
Option	to separate one mole of solid NaCl into one mole of Na+(g) and one mole of Cl-(g) to infinite distance
Option	to convert one mole of solid NaCl into one mole of gaseous NaCl
Option	to convert one mole of gaseous NaCl into one mole of solid NaCl
Answer	2
Solution	
Positive Marks	4
Negative Marks	1

Question	A measure of the stability of the crystal lattice, given by the energy that would be released per mole if atoms, ions or molecules of the crystal were brought together from infinite distances apart to form the lattice is known as
Type	multiple choice
Option	Born-Haber Cycle
Option	Lattice Energy
Option	Hess Law
Option	All of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Lattice energy is inversely proportional to sum of radii of
Type	multiple choice
Option	cation only
Option	anion only
Option	anion and cation
Option	Ions
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The lattice energy of magnesium oxide is more negative than the lattice energy of
Type	multiple choice
Option	oxide ions are larger than fluoride ions
Option	oxide ions are larger than magnesium ions.
Option	oxide ions are more highly charged than fluoride ions.
Option	there is only one oxide ion but two fluoride ions per magnesium ion.
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which among the following is the correct order of tendency to form cation?
Type	multiple choice
Option	$Na^{+} < Mg^{+2} < Al^{+3}$
Option	$Na^{+} > Mg^{+2} > Al^{+3}$
Option	$Mg^{+2} < Na^{+} < Al^{+3}$
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The enthalpy required to dissociated 1 mole of a bonded molecule is called as?
Type	multiple choice
Option	Lattice enthalpy
Option	Bond enthalpy
Option	Ionisation enthalpy
Option	None of these
Answer	2
Solution	
Positive Marks	4

Negative Marks	1
Question	For a compound to be soluble
Type	multiple choice
Option	Hydration energy should be less than lattice energy
Option	Hydration energy should be greater than lattice energy
Option	Hydration energy should be equal to lattice energy
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is the correct order of size of hydrated radii?
Type	multiple choice
Option	$Na^{+}_{(aq)} > Mg^{2+}_{(aq)} > Al^{3+}_{(aq)}$
Option	$Na^{+}_{(aq)} < Mg^{2+}_{(aq)} < Al^{3+}_{(aq)}$
Option	$F^{-}_{(aq)} > Cl^{-}_{(aq)} > Br^{-}_{(aq)} > I^{-}_{(aq)}$
Option	Both (b) and (c)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is correct regarding mobility?
Type	multiple choice
Option	Tendency of ions to move in aq. solution is called mobility
Option	More is the hydration of ion, smaller the mobility
Option	Both A and B
Option	None of these Which is the correct order of mobility of
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is the correct order of mobility of ions?
Type	multiple choice
Option	$Li^{+}_{(aq)} > Na^{+}_{(aq)} > K^{+}_{(aq)}$
Option	$Li^{+}_{(aq)} > K^{+}_{(aq)} > Na^{+}_{(aq)}$
Option	$Li^{+}_{(aq)} < Na^{+}_{(aq)} < K^{+}_{(aq)}$
Option	$Li^{+}_{(aq)} < K^{+}_{(aq)} < Na^{+}_{(aq)}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Electrical conductivity depends on
Type	multiple choice
Option	Ionic mobility
Option	Hydration
Option	Both (a) and (b)
Option	None
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	Which among the following is the correct order of electrical conductivity of ions?
Type	multiple choice
Option	$Na^+ > K^+ > Rb^+$
Option	$Na^+ < K^+ < Rb^+$
Option	$K^+ < Na^+ < Rb^+$
Option	$Na^+ < Rb^+ < K^+$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	An ionic compound is said to be soluble when
Type	multiple choice
Option	ΔH solution = Zero
Option	ΔH solution = -ve
Option	ΔH solution = +ve
Option	All of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Solubility of ionic compound is directly proportional to _____.
Type	multiple choice
Option	Size difference between cation and anion
Option	Only size of cation
Option	Only size of anion
Option	None of these.
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The most covalent halide is:
Type	AlF
Option	$AlCl_3$
Option	$AlBr_3$
Option	AlI_3
Option	4
Answer	
Solution	
Positive Marks	4
Negative Marks	1
Question	LiCl is soluble in organic solvent while NaCl is not because:
Type	multiple choice
Option	Lattice energy of NaCl is less than that of LiCl
Option	Ionisation potential of Li is more than that of Na
Option	Li^+ has more hydration energy than Na^+ ion
Option	LiCl is more covalent compound than that NaCl
Answer	4
Solution	
Positive Marks	4

Negative Marks	1
Question	The correct order of increasing thermal stability of $K_2CO_3, MgCO_3 CaCO_3$ and $BeCO_3$ is
Type	multiple choice
Option	$BeCO_3 < MgCO_3 < CaCO_3 < K_2CO_3$
Option	$MgCO_3 < BeCO_3 < CaCO_3 < K_2CO_3$
Option	$K_2CO_3 < MgCO_3 < CaCO_3 < BeCO_3$
Option	$BeCO_3 < MgCO_3 < K_2CO_3 < CaCO_3$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The most stable carbonate is-
Type	multiple choice
Option	$MgCO_3$
Option	$BeCO_3$
Option	$CaCO_3$
Option	$BaCO_3$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Correct order of covalent character of alkaline earth metal chloride in
Type	multiple choice
Option	$BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2$
Option	$BeCl_2 < CaCl_2 < SrCl_2 < MgCl_2$
Option	$BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2$
Option	$SrCl_2 > BeCl_2 > CaCl_2 > MgCl_2$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the compound is least soluble in water?
Type	multiple choice
Option	AgF
Option	$AgCl$
Option	$AgBr$
Option	AgI
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The M.P. of $SnCl_4$ is less than that of $SnCl_2$, the suitable reason for the observed fact is
Type	multiple choice
Option	There is more charge on Sn^{4+}
Option	The size of Sn^{4+} is small
Option	Ionic potential of Sn^{4+} is high

Option	The shape of $SnCl_4$ is tetrahedral
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of following molecule/species is having maximum number of lone pairs in Lewis-dot structure?
Type	multiple choice
Option	BH_4^-
Option	BF_4^-
Option	CN^-
Option	$COCl_2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Molecule having maximum number of covalent bonds is?
Type	multiple choice
Option	NH_4OH
Option	NH_4Cl
Option	$CO(NH_2)_2$
Option	CH_3OH
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Number of bonded electrons in ethane molecule are?
Type	multiple choice
Option	7
Option	12
Option	10
Option	14
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In which of the following molecule, the central atom has three lone pairs of electrons?
Type	multiple choice
Option	Ammonia
Option	Xenon difluoride
Option	Chlorine trifluoride
Option	Hydrogen sulphide
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The strength of the bonds by 2s-2s, 2p-2p and 2p-2s overlap has the order
Type	multiple choice
Option	$s-s > p-p > p-s$
Option	$s-s > p-s > p-p$

Option	$p-p>p-s>s-s$
Option	$p-p>s-s>p-s$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The shape of CO_2 molecule is similar to:
Type	multiple choice
Option	H_2O
Option	BeF_2
Option	SO_2
Option	None
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In OF_2 molecule, the total number of bond pairs and lone pairs of electrons are respectively:
Type	multiple choice
Option	2,6
Option	2,8
Option	2,10
Option	2,9
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Hypervalent species among the following is:
Type	multiple choice
Option	BF_3
Option	CO_2
Option	SiF_4
Option	XeF_6
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is NOT correct according to the valence bond theory?
Type	multiple choice
Option	The number of unpaired electrons present in an atom is equal to the number of covalent bonds formed by the atom.
Option	Only the atomic orbitals having similar energies are involved in covalent bond formation.
Option	Geometry of the molecule is independent of the directional nature of the overlapping orbital.
Option	Strength of the covalent bond is directly related to the extent of overlapping of atomic orbitals.
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	A hybrid orbital formed from s and p orbital can contribute to
Type	multiple choice
Option	σ-bond only
Option	π-bond only
Option	Either σ or π-bond
Option	Cannot be predicted
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Indicate the wrong statement according to Valence bond theory:
Type	multiple choice
Option	A sigma bond is stronger than π-bond
Option	p-orbitals always have only sidewise overlapping
Option	s-orbitals never form π-bonds
Option	There can be only one sigma bond between two atoms
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	During the complete combustion of methane CH_4, what change in hybridisation does the carbon atom undergo?
Type	multiple choice
Option	sp^3 to sp
Option	sp^2 to sp^3
Option	sp^2 to sp
Option	sp^2 to sp^3
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which orbital is used by oxygen atom to form a sigma bond with other oxygen atom in O_2 molecule?
Type	multiple choice
Option	pure p-orbital
Option	sp^2-hybrid orbital
Option	sp^3-hybrid orbital
Option	sp-hybrid orbital
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following compounds have the same no. of lone pairs with their central atom? I. XeF_5^- II. BrF_3 III. XeF_2
Type	multiple choice
Option	IV and V
Option	I and III
Option	I and II

Option	II, IV and V
Answer	
Solution	
Positive Marks	4
Negative Marks	1
Question	The correct order of hybridization of the central atom in the following species: $NH_3, XeO_2F_2, SeF_4, NO_2^+$
Type	multiple choice
Option	sp^3, sp^3d, sp^3d^2, sp
Option	sp^3, sp^3d, sp^3d, sp
Option	$sp^3, sp^3d^2, sp^3d, sp^2$
Option	sp^2, sp^3d, sp^3d^2, sp
Answer	
Solution	
Positive Marks	4
Negative Marks	1